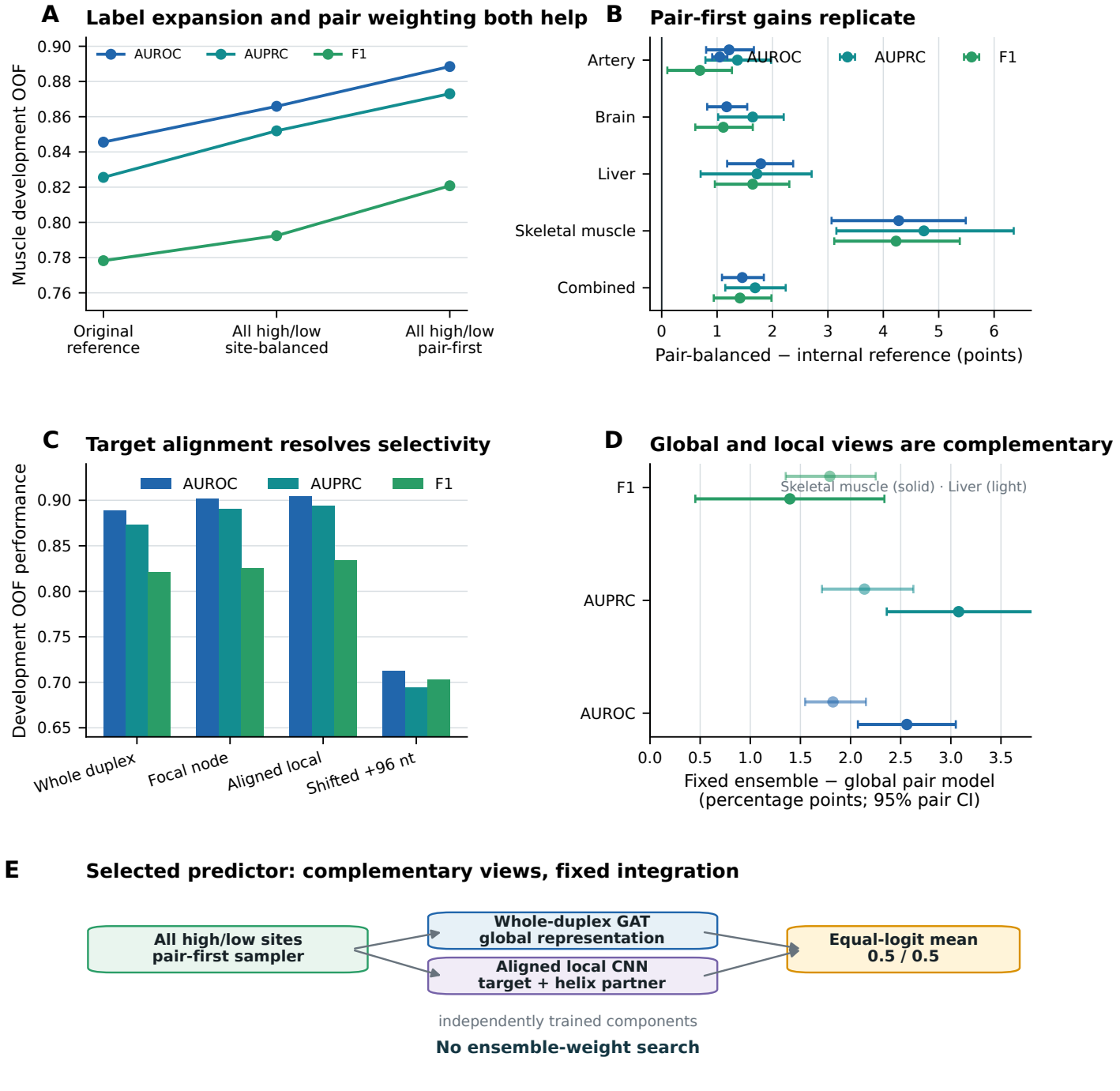
A
Label expansion and pair weighting both help
AUROC
AUPRC
F1
Muscle development OOF
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
Original reference
All high/low site-balanced
All high/low pair-first
B
Pair-first gains replicate
Artery
Brain
Liver
Skeletal muscle
Combined
AUROC
AUPRC
F1
0
1
2
3
4
5
6
Pair-balanced − internal reference (points)
C
Target alignment resolves selectivity
AUROC
AUPRC
F1
Development OOF performance
0.90
0.85
0.80
0.75
0.70
0.65
Whole duplex
Focal node
Aligned local
Shifted +96 nt
D
Global and local views are complementary
Skeletal muscle (solid) · Liver (light)
F1
AUPRC
AUROC
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Fixed ensemble − global pair model
(percentage points; 95% pair CI)
E
Selected predictor: complementary views, fixed integration
All high/low sites pair-first sampler
Whole-duplex GAT global representation
Aligned local CNN target + helix partner
Equal-logit mean 0.5 / 0.5
independently trained components
No ensemble-weight search